

初始化

清除所有定义

```
In[1]:= ClearAll["Global`*"]
```

清除全部

启用下标

```
In[2]:= << Notation`
Symbolize[ParsedBoxWrapper[SubscriptBox["_", "_"]]]
```

下角标框符

数据处理

```
In[4]:= im = {100, 200, 300, 400, 500, 600};
```

```
In[5]:= V1 = {1.03, 2.90, 4.75, 6.60, 8.45, 10.31};
V2 = {-1.10, -2.95, -4.80, -6.66, -8.50, -10.37};
V3 = {2.63, 4.47, 6.34, 8.18, 10.05, 11.90};
V4 = {-2.69, -4.53, -6.39, -8.25, -10.10, -11.95};
```

```
In[9]:= VH =  $\frac{1}{4} \times (\text{Abs}[V_1 + V_3] + \text{Abs}[V_2 + V_4])$ 
```

绝对值

绝对值

```
Out[9]= {1.8625, 3.7125, 5.57, 7.4225, 9.275, 11.1325}
```

```
In[10]:= KH = 167.22;
```

```
In[11]:= is = 6;
```

```
In[12]:= B1 =  $\frac{V_H}{K_H \times i_s}$ 
```

```
Out[12]= {0.00185634, 0.00370022, 0.00555157, 0.00739794, 0.00924431, 0.0110957}
```

```
In[13]:= μ0 = 4 π × 10-7;
```

```
In[14]:= n =  $\frac{3130}{270}$ ;
```

```
In[15]:= B理论 = N[μ0 × n × im]
```

数值运算

```
Out[15]= {0.00145677, 0.00291354, 0.0043703, 0.00582707, 0.00728384, 0.00874061}
```

```
In[52]:= X = {29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15};
```

```
In[16]:= U1 = {8.71, 8.70, 8.69, 8.67, 8.63, 8.60,
              8.54, 8.45, 8.31, 8.07, 7.61, 6.85, 5.44, 3.64, 1.81};
U2 = {-8.73, -8.73, -8.72, -8.70, -8.67, -8.62,
        -8.57, -8.41, -8.34, -8.11, -7.65, -6.90, -5.48, -3.68, -1.85};
U3 = {10.01, 10.01, 10.00, 9.97, 9.95,
        9.91, 9.84, 9.76, 9.62, 9.39, 8.93, 8.18, 6.77, 4.97, 3.15};
U4 = {-10.04, -10.04, -10.03, -10.02, -9.99, -9.95,
        -9.88, -9.80, -9.66, -9.43, -8.96, -8.22, -6.81, -5.02, -3.19};

In[20]:= UH =  $\frac{1}{4} \times (\text{Abs}[U_1 + U_3] + \text{Abs}[U_2 + U_4])$ 
           |绝对值      |绝对值
```

```
Out[20]= {9.3725, 9.37, 9.36, 9.34, 9.31, 9.27, 9.2075,
          9.105, 8.9825, 8.75, 8.2875, 7.5375, 6.125, 4.3275, 2.5}
```

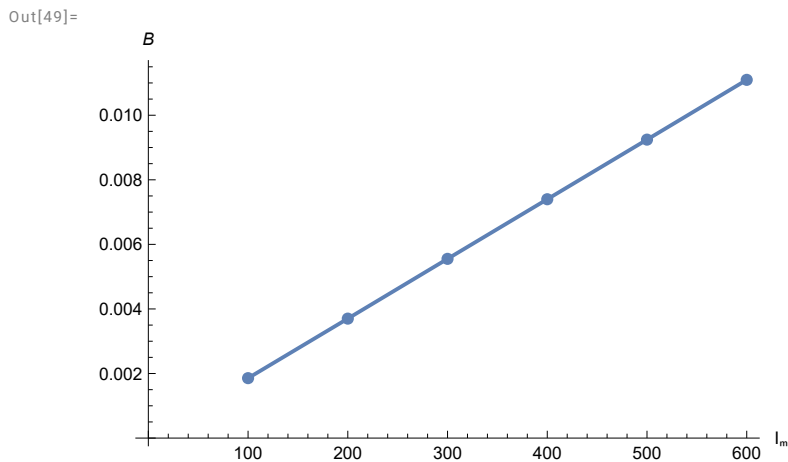
```
In[21]:= is = 5;
```

```
In[22]:= B2 =  $\frac{U_H}{K_H \times i_s}$ 
```

```
Out[22]= {0.0112098, 0.0112068, 0.0111948, 0.0111709, 0.011135, 0.0110872, 0.0110124, 0.0108898,
          0.0107433, 0.0104653, 0.00991209, 0.00901507, 0.00732568, 0.00517582, 0.00299007}
```

绘图

```
In[49]:= ListLinePlot[Transpose[{im, B1}], AxesLabel → {Im, B}, Mesh → All]
           |绘制点集的线条 |转置      |坐标轴标签      |网格 |全部
```



In[54]:= `ListLinePlot[Transpose[{X, B2}], AxesLabel → {HoldForm[X], B}, Mesh → All]`
[绘制点集的线条](#) [转置](#) [坐标轴标签](#) [保持表达式](#) [网格](#) [全部](#)

Out[54]=

